Siddhant Kushwaha

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Work Experience

Cohesity
Software Engineer 3 August 2024 - Present

- Built a machine-learning based document classification service that scans backup data, extracts text from files, and tags them for a data catalog, running model inference on CPU-only compute nodes, using **Go**, **Python**, and **Apache Tika**.
- Designed a configurable tagging pipeline with pluggable ML classifiers, streaming data transfer over **gRPC**, and health checks for reliable deployment on **Kubernetes**.
- Developed clone-based recovery support for Oracle CDB databases on Linux and AIX, improving restore speed by **50%** compared to traditional methods.
- Extended Cohesity's data protection platform to back up and recover Oracle databases running on **AWS**, extending support to customers hosting Oracle in the cloud.

Commvault
Senior Software Development Engineer January 2020 - July 2024

- Played a pivotal role in designing, developing, and implementing solutions that facilitated seamless backup and indexing of a staggering **12 billion objects** distributed across **45,000 laptops** for a customer using the **Apache SOLR** framework.
- Designed a solution that facilitates the indexing of files larger than 1GB, stored in segments, to enable efficient data validation, as well as browsing and restoration of fully backed-up files.
- Implemented best practices for backing up indexes and retention strategies, ensuring data availability in case of index-related issues, and reducing downtime during data restoration processes, resulting in space savings of over 50%.
- Wrote an enhanced cache cleanup strategy for index transactions, reducing disk space usage by transaction logs by over 80%.

Education

Indian Institute of Information Technology, Sri City GPA: 8.22
B.Tech Computer Science and Engineering August 2016 - May 2020

Aryaman Vikram Birla Institute of Learning Class XII *Percentage: 92.8%*
Haldwani, Uttarakhand Class X *GPA: 9.8*

Projects

- Developed and launched **ProText**, an Android SMS app that utilizes on-device intelligence via TensorFlow-Lite models for SMS classification, helping in inbox organization and spam notification prevention. It has garnered over 100 downloads to date.
- Built an email management tool that periodically labels and unsubscribes emails based on customizable rules, significantly boosting productivity. Received **17 stars** on **GitHub** for this project.

Achievements

- Secured second place at **Google India Hackathon 2018** by developing an app using an on-device deep learning model to accurately detect potholes, enhancing road safety and infrastructure maintenance.
- **Winner, Bot Fight Competition 2023 at Commvault**: Wrote a winning bot for the game of Blokus, showcasing advanced strategic algorithms and programming skills.
- **Runner Up, Commvault Create 2024**: Leveraged large language models (LLM) and retrieval augmented generation (RAG) to bring conversational search and insights to enterprise data.

Skills

- Programming Languages: C, C++, Go, Java, Kotlin, Python, JavaScript
- Fundamentals: Algorithms, Object Oriented Design, Databases
- Tools/Libraries/Frameworks: Git, MySQL, MS SQL, MongoDB, Apache SOLR, Apache Tika, gRPC, Kubernetes, Android SDK, Django, Spring